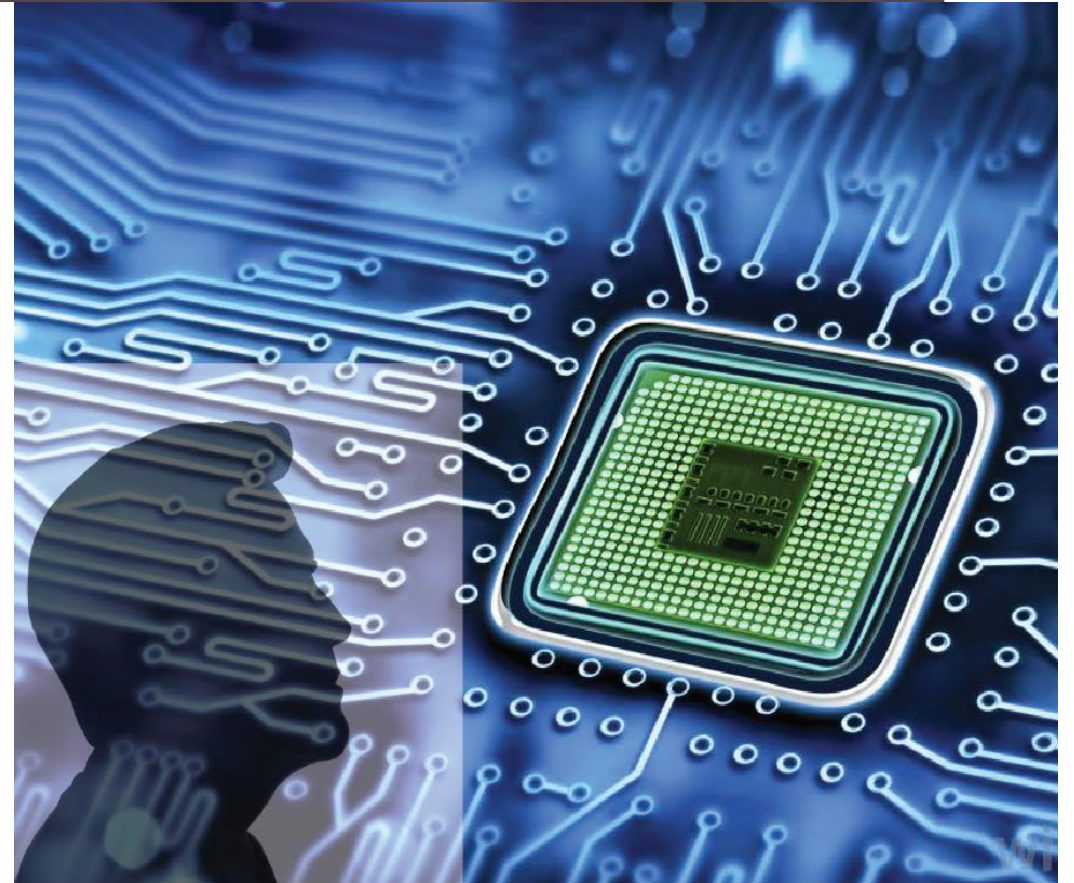


VLSI DESIGN FLOW

Week-14 Lecture-2 Physical Design

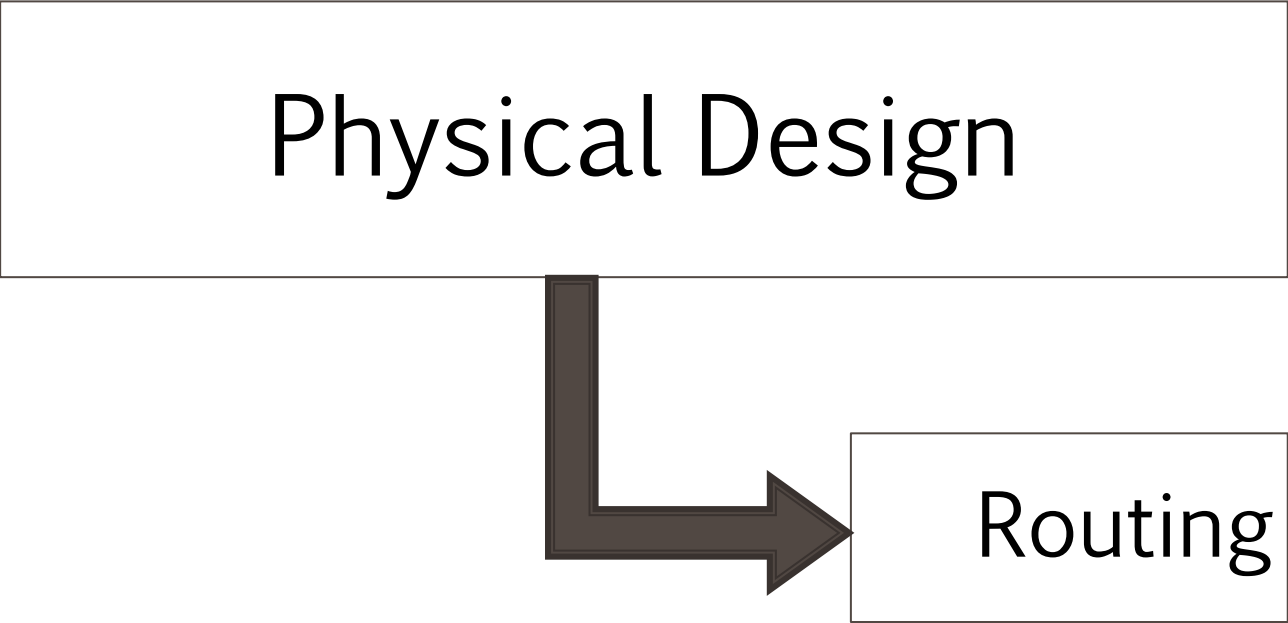
Sneh Saurabh
27th November, 2024



VDF: Revision/Announcements

-





```
graph TD; A[Physical Design] --> B[Routing]
```

Physical Design

Routing

Global Routing: Routing Model

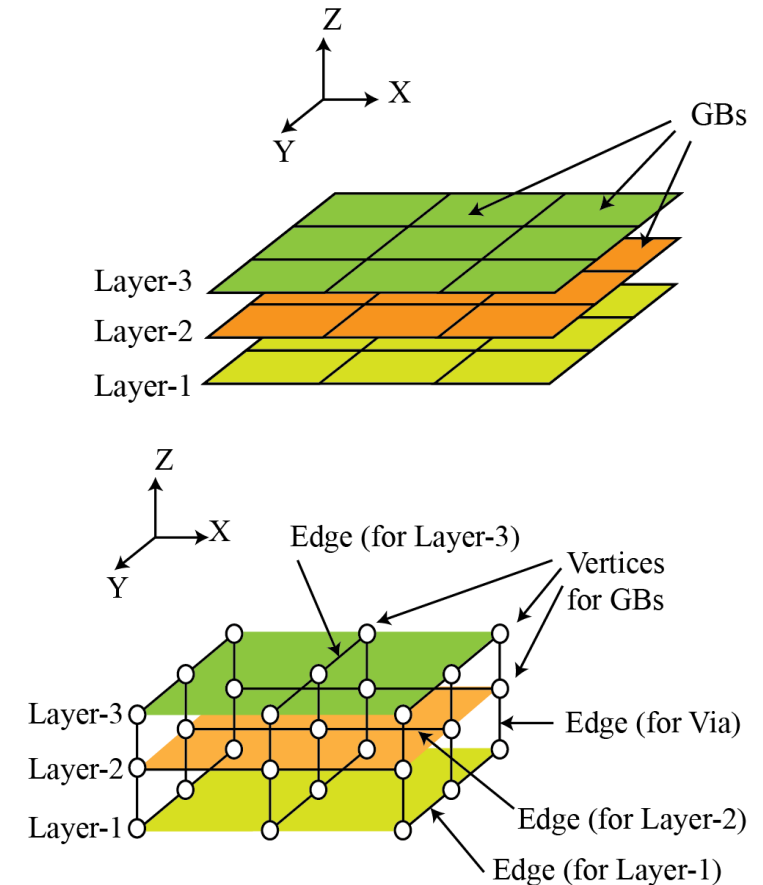
- Routing area is divided into rectangular grids
- Each rectangular region is called **Global Bin** (GB or global tile, routing tile global cell, or bucket)

Grid graph is built using GBs:

- Vertices (v): GBs
- Edges (e): represent boundary between adjacent GBs
 - Create edges between two adjacent vertices only if they lie along the preferred direction of routing
 - Create edges between two vertices that lie vertically adjacent (for vias)

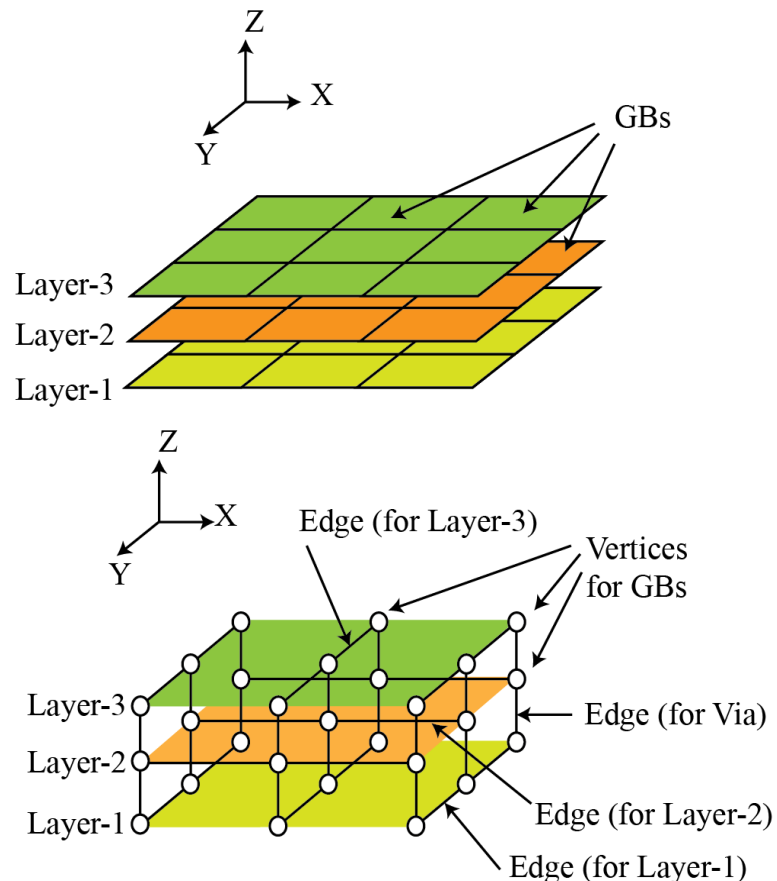
Determining a route for a net:

- Associate a pin with a vertex in the grid graph using the GB in which the pin is placed
- Find an *optimal* path in the grid graph that contains all the vertices for that net



Global Routing: Routing Model

- When a net crosses the boundary of two GBs
 - Net utilized the corresponding edge



- **$USE(e)$** : number of nets utilized for an edge
 - Also called ***demand*** for an edge
- **$CAP(e)$** : quantifies availability of routing resources for an edge
 - Also called ***supply*** for an edge
 - Routing blockages limit the capacity
 - Depends on layer and design rules

Capacity constraint: $USE(e) \leq CAP(e)$

Overflow $OF(e) = USE(e) - CAP(e)$

Congestion $CG(e) = \frac{USE(e)}{CAP(e)}$

Global routing attempts to route all the nets such that $OF(e) = 0$

Global Routing: Challenges

Runtime versus Accuracy

- Runtime improvement:
 - Increased size of GBs
- Accuracy:
 - Anticipate problems that will be encountered by detailed routing (ease design closure)
 - Better modelling of routing resources and design rules

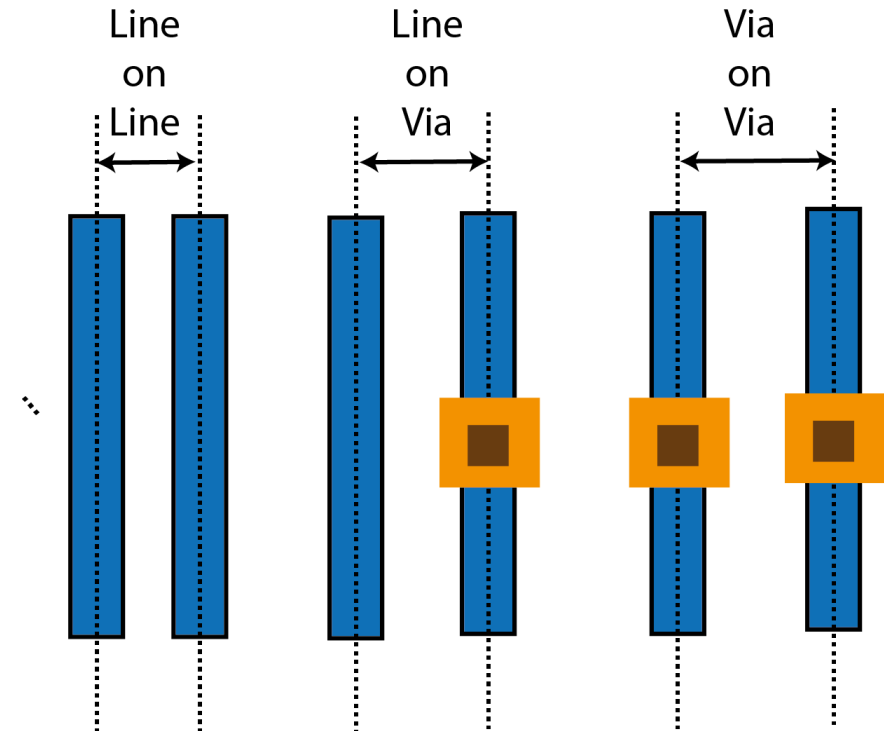
Detailed Routing

Detailed Routing: Goals and Constraints

Goal: To determine the exact layout of each net, including all the attributes of wire segments such as width and location

Constraints and Considerations:

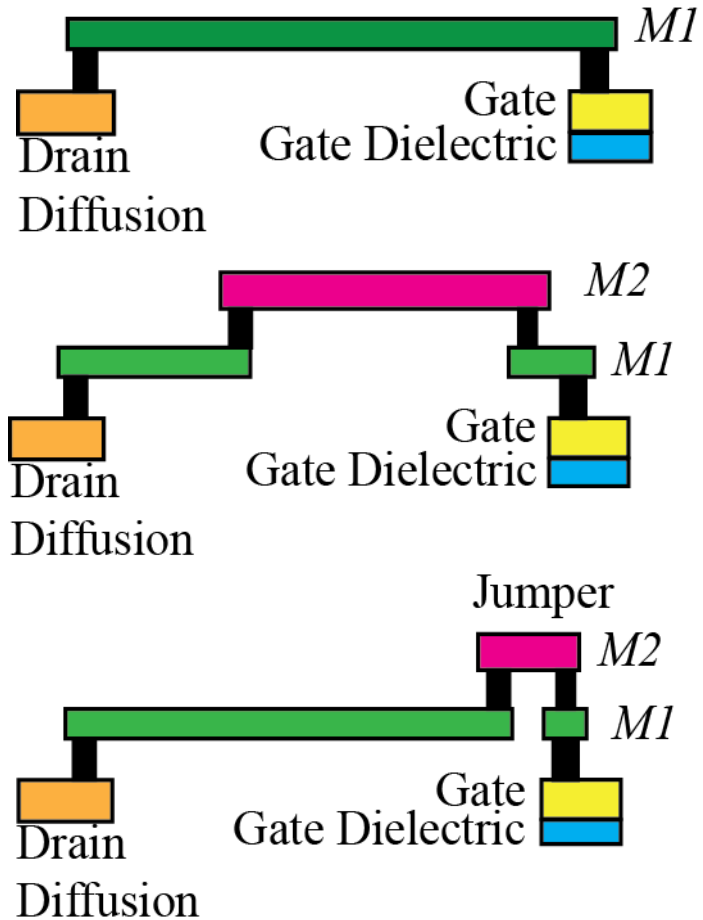
- Connectivity, design rules
- Timing, Signal Integrity, Runtime
- More number of metal layers available
 - Over-the-Cell (OTC) routing done
- Performed using a **detailed routing grid**
 - Tracks with uniform spacing (routing pitch)
 - Allows easy automation
- Routing pitch is defined as minimum spacing allowed in a technology (design rules):
 - Line on Line: can be too aggressive (DRCs)
 - Via on Via: can be too conservative
 - Line on Via: trade-off



Routing: Final Optimization

- Timing analysis with interconnect layout and parasitics taken into consideration
- Improve timing issues by:
 - Gate upsizing, buffering, automatic/manual re-routing and wire widening
- Reduce power consumption: gate downsizing, removing buffers
 - Cell change with similar *footprint*
- Fix signal integrity issues
 - Increasing distance between wires

Antenna Effect



- Prone to antenna effect-induced gate damage

- Avoid long M1 line directly connected to the gate terminal
 - Instead use higher layer metal (M2) for routing

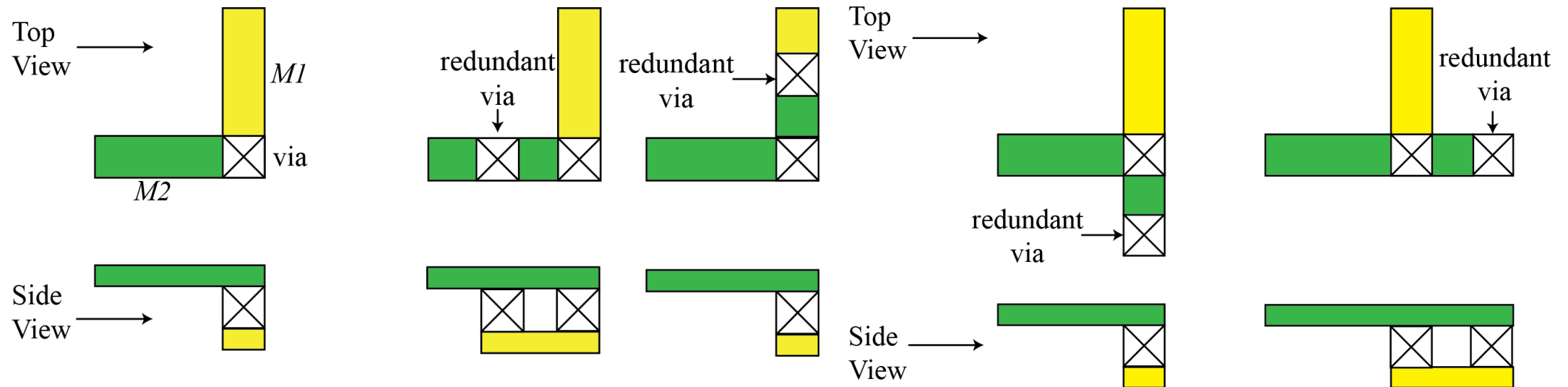
- Add a jumper to fix antenna rule violation after routing

Reliability Issues: Via Defects

Vias are prone to failure

- Can lead to open circuit defects
- Yield loss and reliability problem

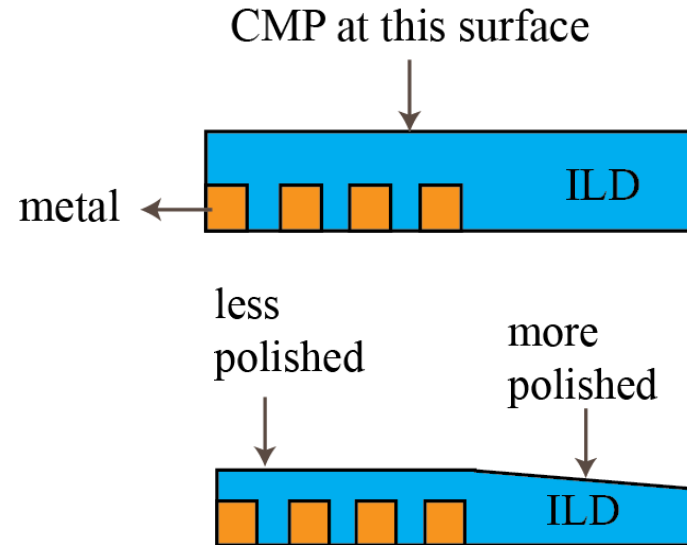
Solution: add redundant vias



Manufacturability Issues

- Difference in hardness of metal and dielectric, CMP can produce **irregular topography**

- Irregular topography can lead to yield loss



- Add dummy *metal fills* during routing to ensure more uniform metal density
- The existence of dummy metal fills in the vicinity of current-carrying wires can affect the coupling capacitance.
 - Dummy metal fills can impact the timing of a circuit.

Layout Extraction

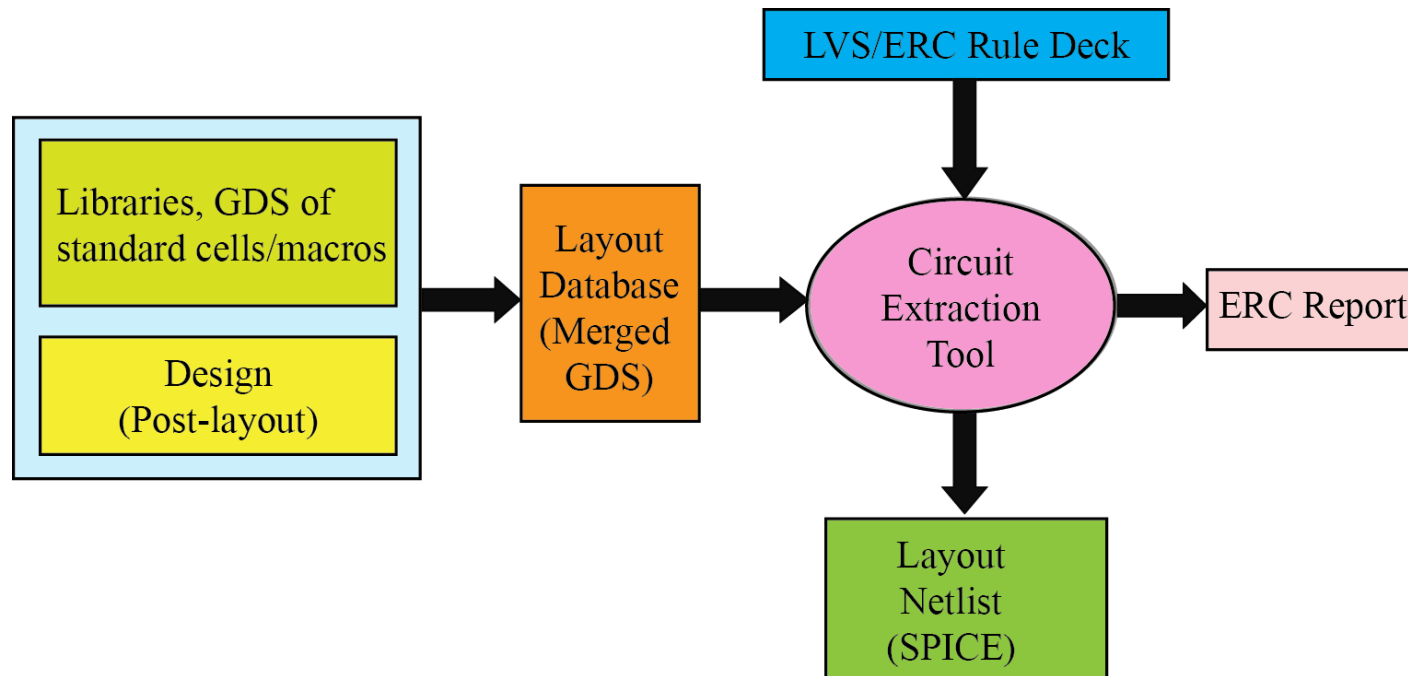
Layout Extraction: Basics

- Layout describes the shapes required on each fabricated layer
 - Difficult for other post-layout verification tools to work with this information
- Extract various information from the layout
 - Problem for a verification tool is greatly simplified

Layout extraction consists of two major tasks:

1. **Circuit Extraction:** devices and interconnections
2. **Parasitic Extraction:** parasitic resistance, capacitance and inductance

Circuit Extraction



Inputs:

- Merged GDS: layout of the design and standard cells/macros
- Extraction rules: for devices and connections (comes from foundry)

Outputs:

- Layout Netlist: typically in SPICE
- ERC Report

Parasitic Extraction

- **Resistance:** each net may be divided into multiple net segments
- **Capacitance:** need to account for various components

Technology pre-characterization

- Performed once for a given technology
- Enumerate millions of sample geometries/structures
- Use accurate Field Solvers for computation
 - Values stored in lookup tables or empirical formulae created using curve-fitting
- Highly time consuming

Chip-level parasitic extraction:

1. Technology pre-characterization
2. Pattern Matching

Pattern Matching

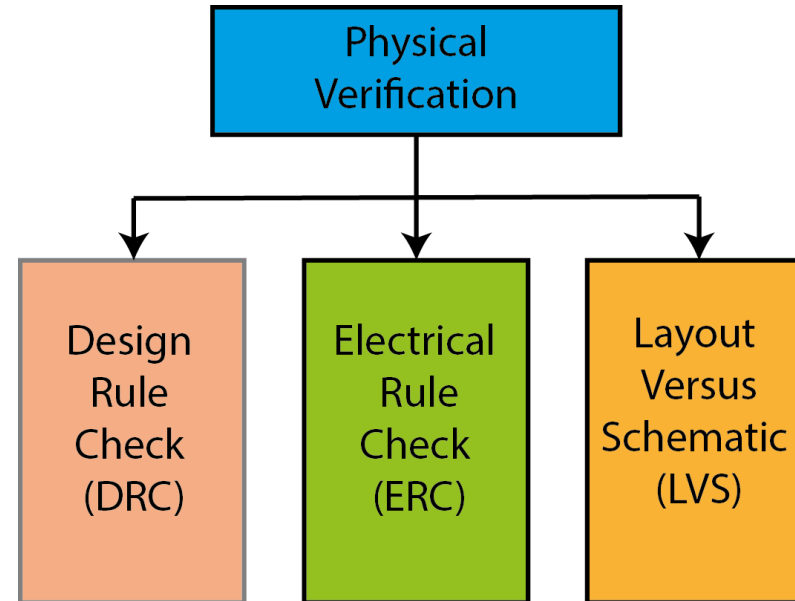
- Partition a layout into smaller windows
- Match windows with pre-characterized patterns
- Compute the capacitance with the help of lookup tables or empirical formulae
 - Actual geometries of the layout used

Inductance:

- More challenging to extract
- Fortunately, can ignore for most nets
- Numerical techniques used

Physical Verification

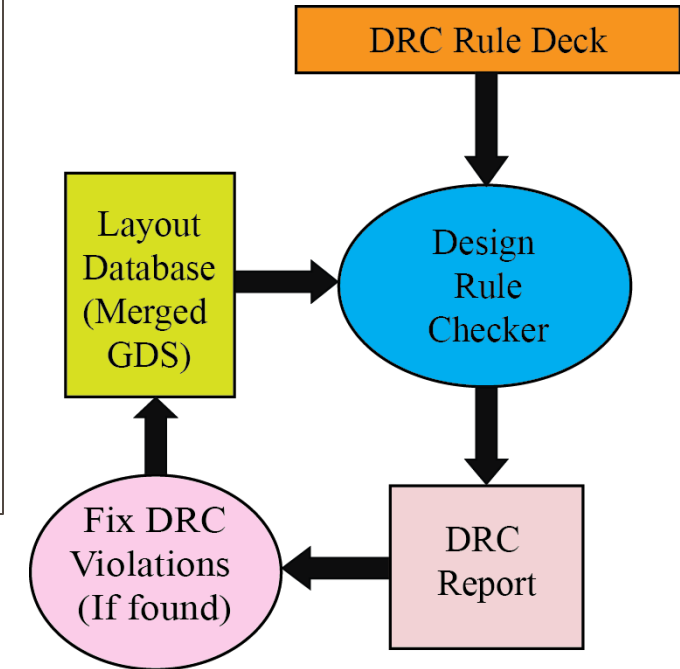
Physical Verification



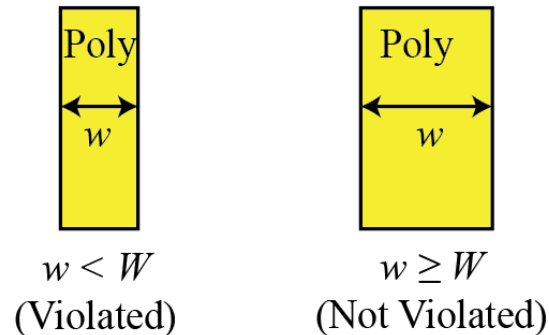
Physical Verification : DRC

Design Rule Check (DRC):

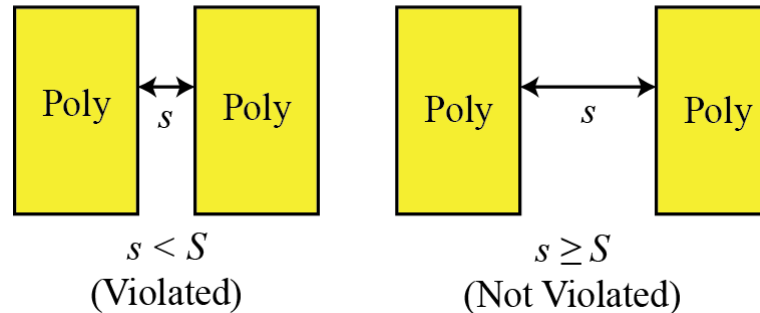
- Ensure that the layout meets the constraints required for manufacturing
- Rules are defined by the respective foundry
 - Achieve a good yield and improve reliability
 - Vary with the technology
 - Become more complicated with advancement in technologies



Rule: Minimum Width = W



Rule: Minimum Spacing = S



Physical Verification : ERC

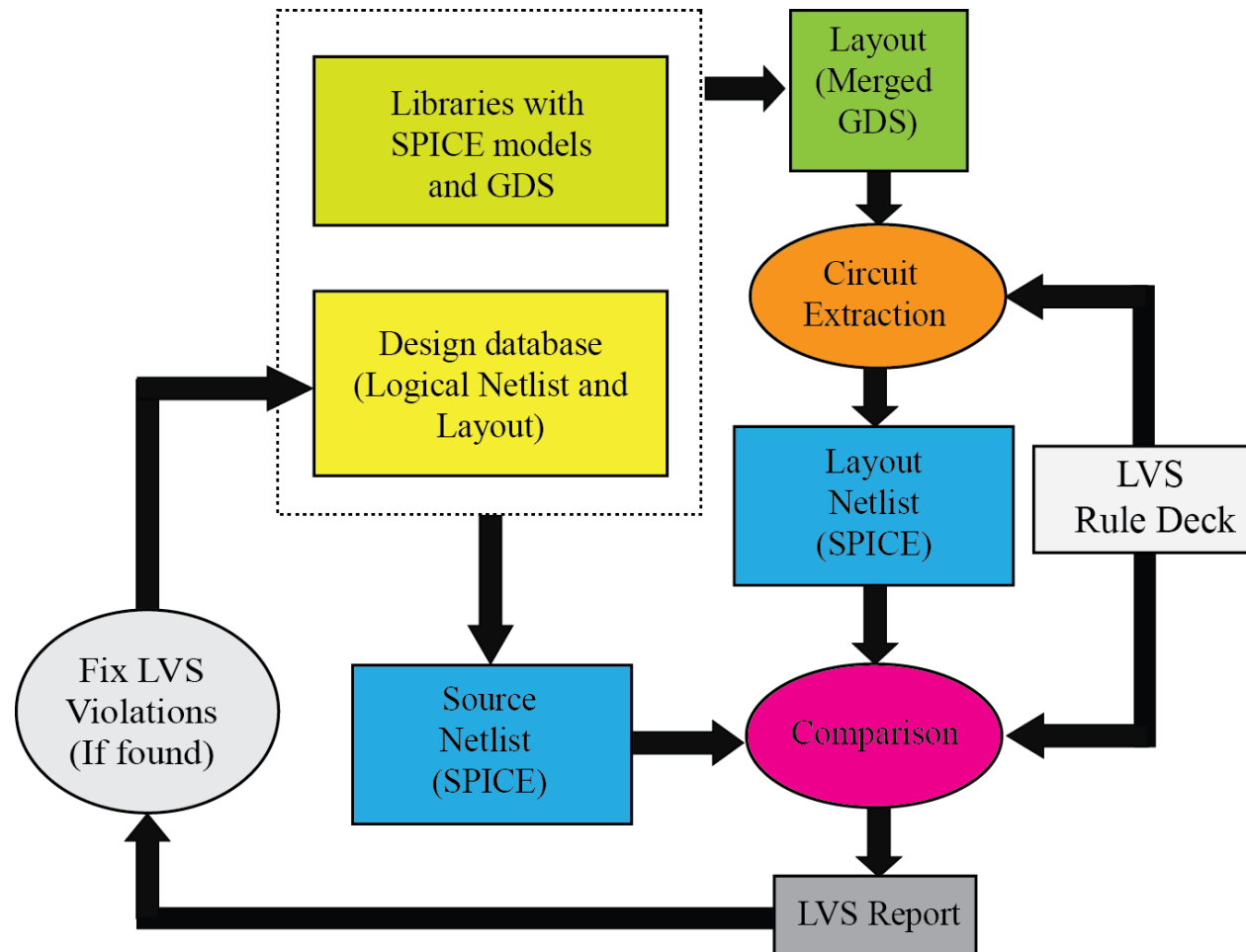
Electrical Rule Check (ERC):

- Check design for electrical connections that can be problematic
- Examples: Shorts/Open, Floating Gate, Floating Nets etc.

Physical Verification : LVS

Layout versus Schematic (LVS):

- Verifies whether the layout corresponds to the original schematic (netlist) of the design



Compares:

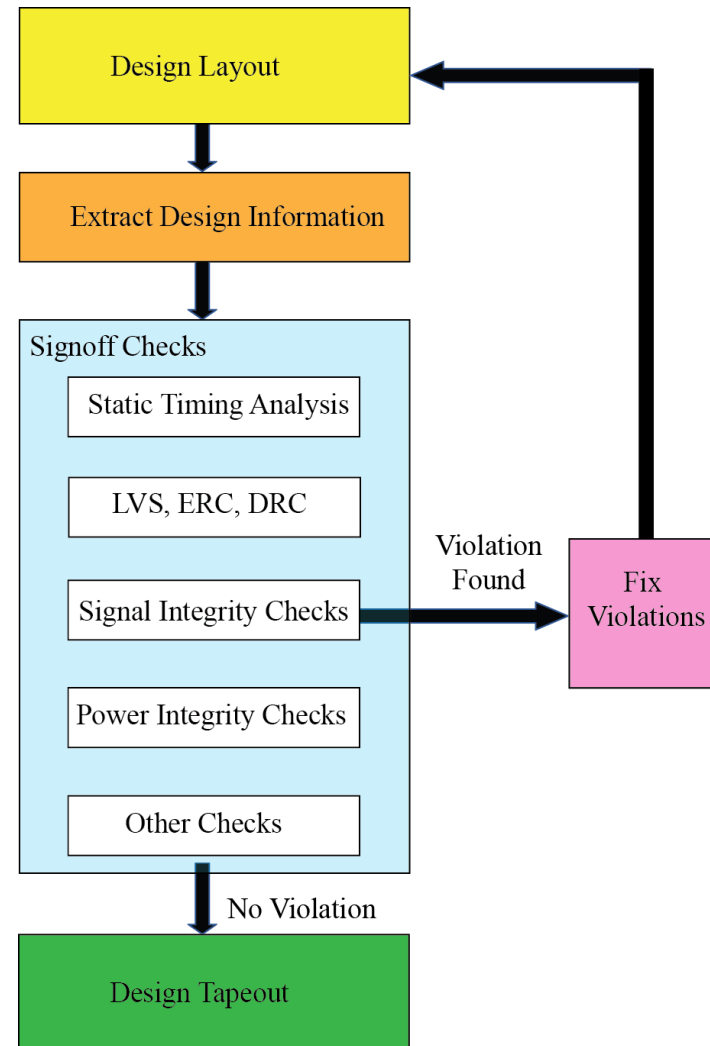
- Layout netlist (extracted)
- Source netlist (schematic): logical netlist combined with device information

Signoff

Signoff

Signoff:

- Series of verification steps that must be carried out before sending the layout (GDS) to a foundry
- Ensures that the layout delivers the intended functionality and meets various figures of merit



Engineering Change Order (ECO)

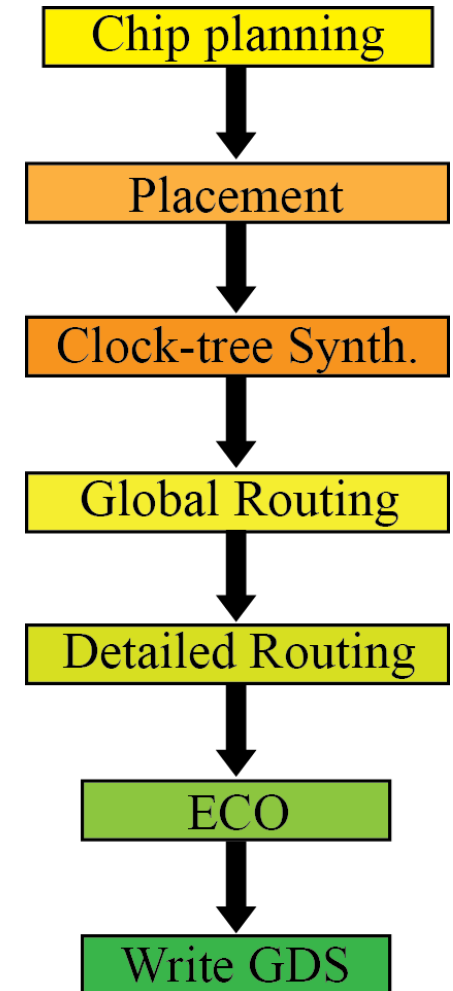
- Sometimes last moment incremental changes in a design needed
 - Fix issues discovered late or new incremental functionality
- Risky changes
 - Incorporate them using Engineering Change Order (ECO)

ECO tools:

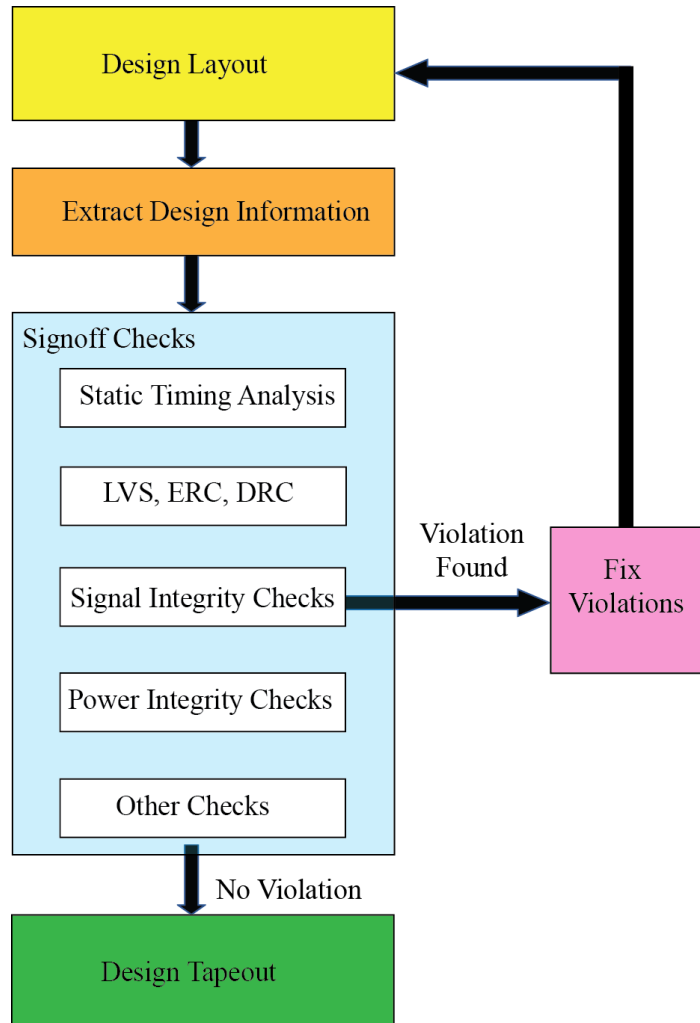
- Enable making targeted incremental changes, rather than re-implementing the entire design
- Verify the correctness of the ECO changes.
 - Saves designer time, effort, cost, and risk

Types of ECO changes

- Functional ECO: changes logic
 - Logic re-synthesis or use spare cells
- Direct changes in layout to fix setup/hold time violations, SI-related issues, and design rule violations.



Signoff and Tapeout



- Design Tapeout: send the GDS for fabrication
- Occasion to celebrate !

References

- S. Saurabh, “Introduction to VLSI Design Flow”. Cambridge: *Cambridge University Press*, 2023.

Wishing you a bright future!